

Flying Saucers-Superconducting Whirls Of Plasma

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In the following article you will find one of the best-kept secrets of modern times - and with its publication, it no longer remains a secret. This secret is the answer to the question most asked of the editors of FLYING SAUCERS: "What are the flying saucers?" Here is the answer, but only in part. The reader is warned that what he is about to read may be technical in its language, but the editors believe it is stated simply enough to be understandable to the readers of FLYING SAUCERS. To physicists (who are referred to the bibliography), the facts contained in this article will be obvious and unchallengeable. In future issues, FLYING SAUCERS intends to continue this "definition" of flying saucers, and prove that the editors have been correct from the beginning in insisting the UFO is native to this planet, and not from interplanetary space. The editors invite scientists the world over to examine the facts presented here, to engage in further research, and to aid us in placing the riddle of the UFO where it belongs - in the realm of scientific reality, backed up by the scientific method. The editors of FLYING SAUCERS believe they are making history with this article, and with future enlargement of the demonstrable facts and theories contained therein. After nineteen years of mystery, a break-through has been made!

During recent years research on magnetohydrodynamics has provided humanity with a great number of discoveries of new physical phenomena. It has not been possible to predict these phenomena. We find ourselves at the beginning of a startling development, whose consequences are of immense importance for the future of humanity on planet Earth.

When a plasma is compressed by very strong magnetic fields, it has a tendency to form special shapes (plasmoids). The shapes assumed by plasmoids are unexpectedly rich and have provided information considerably in advance of the ability of theory to predict. A commonly observed hydromagnetic property of plasma is the tendency of the plasma spontaneously to form shreds or thin filaments. These filaments often form spirals in the container or magnetic bottles in which the plasma is accelerated.

When a plasma with high electrical potential is compressed by a very strong magnetic field, and when at the same time a very strong magnetic field is created inside and along the axis of the column of plasma, the particles of the plasma will move in tight spirals around and along the magnetic lines of force. Associated with this corkscrew movement are

two important qualities: the number of revolutions per plasma density is so high that the number of distant-encounter collisions per second exceeds the gyro-frequency, the cork-cut takes place, so to speak, transforming the extremely tight spirals into a great number of ring-shaped plasma whirls. These ring-shaped plasma whirls are superconducting. As the electrical potential in them is enormous and now does not encounter any resistance, very powerful hydromagnetic shock waves are emitted.

For many years superconductors were thought of simply as substances whose electrical resistance vanishes below a certain temperature. This concept met with a number of minor but significant difficulties, which eventually led to the realization that superconductivity represents a state of matter in which a large number of electrons are in a coherent state of motion. Such a state can persist indefinitely in flagrant violation of the classical law that a circulating electron must lose energy by radiation.

A typical property of superconductors is their impenetrability for magnetism. As the ring-shaped whirls of plasma are superconducting, magnetic fields cannot penetrate them, and they are actually im-

prisoned in their own magnetic fields (and those of the other neighboring whirls). That is why a ring-shaped current can remain stable indefinitely and at the same time emit powerful hydrodynamic shock waves without a supply of energy. These ring currents of superconducting plasma present certain analogies with the ring currents created in superfluid helium (below 2.2 degrees K) when bombarded with alpha particles, and with smoke rings.

Because of these dramatic macroscopic quantum effects exhibited by the superconducting ring currents, they can under certain conditions pave their way through the containers and escape from the laboratories without interruption of their coherent state of motion. They will then follow magnetic lines of force.

In a separate paper I have mentioned that such ring currents are created in the magnetosphere due to the interaction of ionized particle waves from the sun and the geomagnetic field. Earth's magnetic lines of force trap the waves of charged particles and cause them to move in spirals. Under certain conditions the spirals become so tight that huge stable superconducting ring currents are created. When the superconducting ring currents con-

sist of trapped low-energy protons, they cause electromagnetic radiation at frequencies of about one cycle per second. And ring currents of trapped electrons give frequencies in the audible range, which is from 20 cycles to 15,000 cycles per second. The luminous phenomena are also called "HASER" an acronym standing for Hydromagnetic Amplification by Stimulated Emission of Radiation. It is to be remembered that the particles are in a coherent state of motion in the ring currents.

A great number of ring currents are often combined in huge luminous cigar-shaped fields, but they can also separate and appear as smaller oval shapes or quite small luminous balls. Due to perturbations of the magnetic lines of force they can move nearer to the surface of Earth. Such strange bright lights (Unidentified Flying Objects) have been seen by millions of people. The shapes reported correspond very well to the shapes of superconducting ring currents. As they are very sensitive to magnetic lines of force, they have often been reported hovering over or moving along powerful electric current wires. When following the magnetic lines of force of Earth, the path will look like a zigzag line. They will also tend to move along certain geographic lines because of deviations of the geomagnetic field due to deposits of magnetic material in the Earth (orthotheny).

UFOs are often changing color when changing speed. In the superconducting ring currents the light is emitted by electrons that exhibit coherent wave properties over macroscopic dimensions. And a magnetic field is capable of varying the energy of quantum levels and hence the frequency of the emitted light, as well as reducing drastically the threshold for the onset of laser action.

Many other strange phenomena accompany the UFOs. The strong hydromagnetism gives magneto-optic reflection and absorption, so that the ring currents can look solid and metallic. Also magneto-acoustic effects are common. In the superconducting ring currents the electrons are in a coherent state of motion thus producing coherent electromagnetic waves as well as hydrodynamic shock waves are cap-

able of giving rise to compression and rarefaction of particles. The free electrons in a current will thus change their density. Now, the behavior of electrons in a metal follows the rules of Fermi statistics, according to which the energy of electrons is always related to their density. Thus when the density of the electrons in a metal is disturbed, the distribution of their energies is also disturbed. Electric currents will be stopped causing black-out of electric powered machines and instruments. To re-establish the equilibrium distribution a certain relaxation time is required. Then the electric powered devices will re-assume their function as if nothing had happened. Such black-outs are some of the most reported effects during close approach of UFOs.

Two theories have been put forward in order to explain the enormous amounts of energies released by hydrodynamic plasma. The first one says that there are hidden or uncounted extra positive and negative charges and hidden or uncounted magnetic movement in the plasma. The second theory says that the energies are released when the hydro-magnetic whirls of plasma assume the same geometric dimensions as the electrohydromagnetic threads appropriate for an elementary particle. The Mach principle taken into consideration, the self energy of any elementary particle must be described within the framework of the universe as a whole. The universe is considered as a transient stage in which a transient latent universal potential is transformed into quantized active local energies by processes inside elementary particles. Now when we know that the energies are released during a superconducting state of plasma, the first theory can be ruled out definitely. The second theory is still valid. The superconducting state of electrohydromagnetic threads in particles and of coherent whirls of elementary particles present us only with the information on how infinite amounts of energies can be released, but it does not provide us with any information on why and from where the energies are coming. Any hypothesis saying that the superconductivity in itself possesses self-energy infinites must be considered as irrelevant. The theory on a transient

latent universal potential has considerable conceptual merit in as much as it also conforms to cosmologic concepts and to universal constants.

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